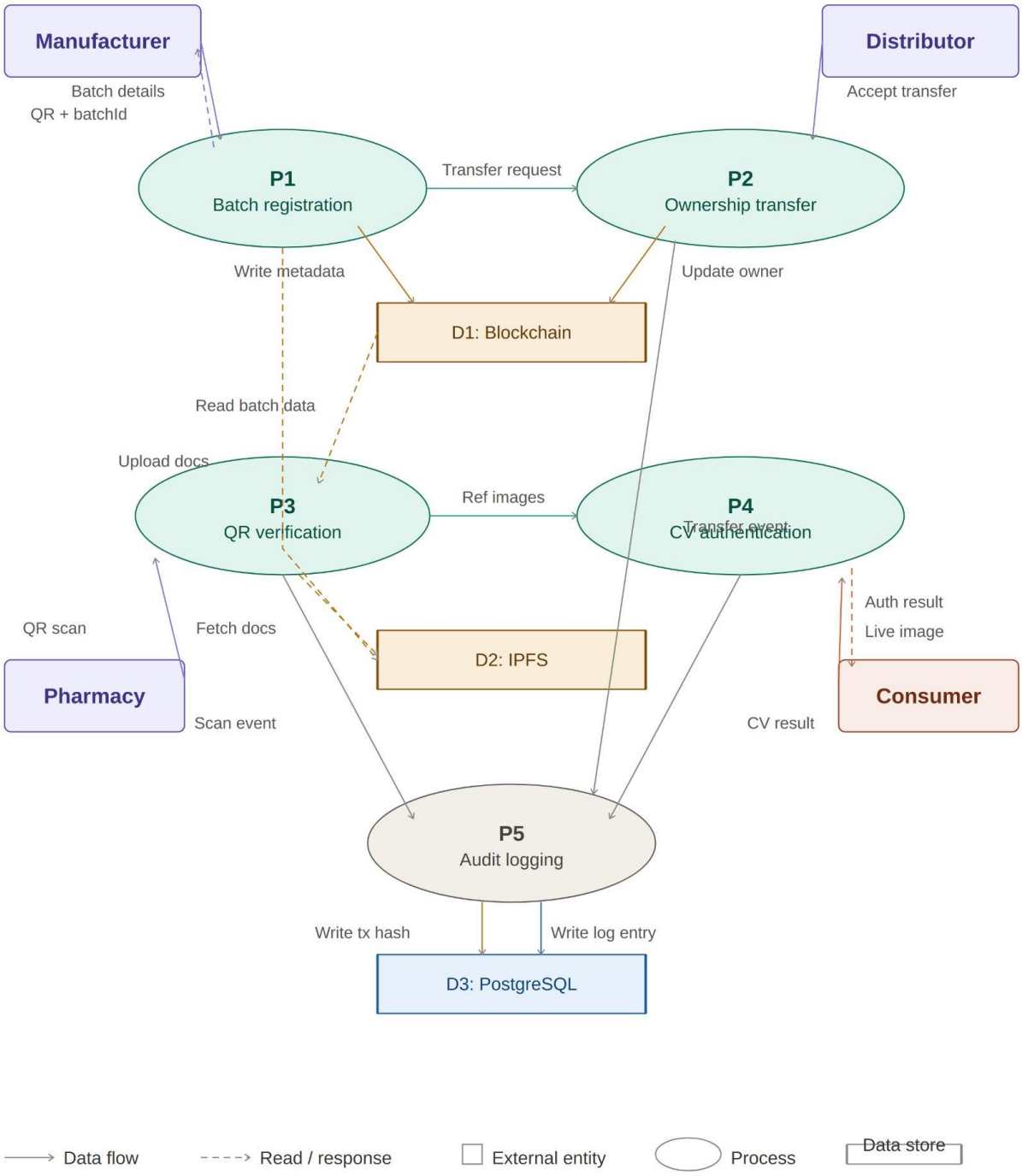
MedSecure – Data Flow Diagram (Level 1)



Description

The Level 1 DFD decomposes the single MedSecure process from the Context Diagram into five distinct sub-processes, revealing internal data flows, data store interactions, and inter-process communication.

P1 – Batch Registration: Receives batch details from the Manufacturer, writes metadata to the Blockchain data store (D1), uploads documents to IPFS (D2), and issues a QR code + batchId back to the Manufacturer. It also emits a transfer request to P2.

P2 – Ownership Transfer: Accepts the transfer request from P1 and the Accept Transfer signal from the Distributor. Writes updated owner records to D1 (Blockchain).

P3 – QR Verification: Receives QR scan events from the Pharmacy, fetches on-chain batch data from D1, retrieves documents from D2 (IPFS), and passes reference images to P4. It also forwards events to P5 for audit logging.

P4 – CV Authentication: Receives reference images from P3 and a live packaging image from the Consumer. Performs visual comparison and returns the authentication result (Auth result) to the Consumer. Sends its CV result to P5.

P5 – Audit Logging: Aggregates events from P3 and P4 and writes both a transaction hash and a structured log entry to D3 (PostgreSQL), forming the persistent audit trail.

Three data stores underpin the system: D1 (Blockchain) for immutable on-chain records, D2 (IPFS) for decentralised document and image storage, and D3 (PostgreSQL) for structured off-chain audit logs.